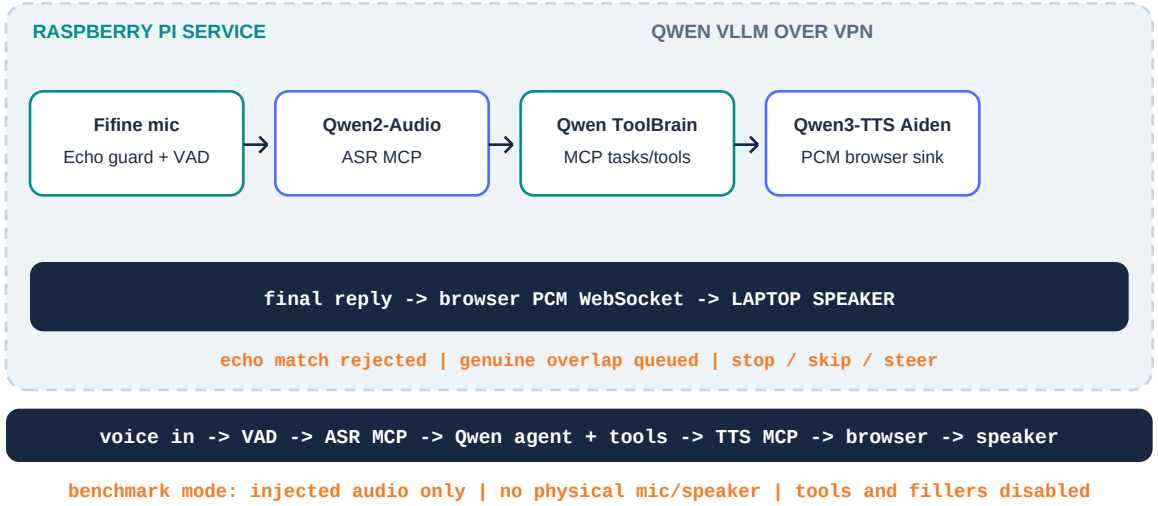
# Full-Duplex Voice Agent on a Raspberry Pi



## 1 SYSTEM DIAGRAM

*input -> Pi -> agent -> laptop speaker*




- Fifine mic is captured by the Pi service; laptop browser is the speaker sink in the intended demo route.
- Pi owns echo guard, Silero VAD, MCP servers, task safety, and duplex state; Qwen services are reached over VPN.
- Qwen2-Audio ASR -> Qwen ToolBrain / MCP -> Qwen3-TTS Aiden; output is 16-kHz PCM to the browser.
- The agent supports web/file/build tools and confirmation-gated channels; no model training is required.
- Protected overlap keeps listening open: matched echo is rejected; genuine speech is queued; stop, skip, and steer are explicit controls.

## 2 PROMPT

*versioned in config/config.yaml*

**SYSTEM PROMPT (QWEN FAST MODE) - EXCERPT**

```
Filipino / English / Taglish voice assistant for a Philippines customer-support pilot.
Formal, respectful, concise spoken replies; never flirtatious or suggestive.
Use bounded project tools when asked; confirm calls, SMS, Telegram, or email first.
Never expose credentials or claim an external action succeeded without tool evidence.
For long work: "I am processing your request now."
```

## 3 CONTEXT ENGINEERING

*short, safe, observable turns*

**GOES IN:**

- system persona + PH language policy
- ASR transcript and VAD endpoint
- rolling history + tool results
- overlap, echo, and control events

**MANAGED:**

- fast mode; hidden thinking off
- 24-round tool budget + safe summary
- 1-3 spoken sentences; short history
- latest-prompt queue + background status
- sanitized terminal; no credentials or CoT

# MCP Services, Evaluation & Model Serving

*build + run owned MCP services on the RPi*

## 4 MCP SERVICES *+ model serving: one table, one truth*

| Component | MCP server / transport | Tools | Model | Served where / notes |
|---|---|---|---|---|
| ASR | me3-asr / FastMCP | transcribe(audio) -> text | Qwen2-Audio-7B-Instruct | Pi service; Qwen vLLM over VPN; final at VAD endpoint |
| TTS | me3-tts / FastMCP | speak(text) -> audio | Qwen3-TTS-12Hz-1.7B-CustomVoice; Aiden | Pi service; Qwen vLLM over VPN; one formal voice |
| LLM | ToolBrain / MCP client | function calling | Qwen/Qwen3.8-27B | Remote vLLM; fast_mode=true; hidden thinking=false |
| VAD | me3-vad / FastMCP | detect_speech(frames) | Silero ONNX | On-device Pi CPU; 16 kHz mono; endpointing |
| Task + channels | me3-tasks / FastMCP + REST | files, web, build, timers, ME2 parity, channels | n/a / provider APIs | Pi-owned tools; live IoT/provider effects require confirmation |

## 5 EVALUATION & BENCHMARK *updated class direct-feed protocol*

- One clock: benchmark direct feed uses the assistant machine clock and records stage marks per turn.
- Injection only: no physical microphone or speaker; no acoustic feedback in the latency run.
- Benchmark mode: no tool schemas, external agent/harness calls, fillers, backchannels, or real channel side effects.
- Report mean + p95 over the updated stage-latency phrases and preserve JSON, CSV, HTML, and latency chart artifacts.

| Metric | Definition | Target | Mean | p95 |
|---|---|---|---|---|
| E2E latency | end of speech -> audio out | 4.0 s | | |
| ASR final | VAD end -> transcript | 0.8 s | | |
| LLM TTFT | transcript -> first token | 1.5 s | | |
| TTS TTFB | text -> first audio chunk | 0.6 s | | |
| Barge-in | speech -> playback stopped | 0.8 s | | |
| WER | ASR accuracy on test set | 15% | | |

### DEMO (deck 2)

Available now: injected Qwen voice/tool turn, web-app creation, and ME2 parity tools for music, weather, media, IoT state, timers, reminders, and channels.

The intended physical demo is Fifine mic on Pi -> Pi ME3 -> browser -> laptop speaker.

Not claimed here: a clean human Pi spoken turn and physical speaker observation.

### RESULTS (deck 2)

Workstation acceptance: 70 tests passed; all four MCP health checks passed; Qwen injected voice/tool path passed with zero errors.

Qwen fixture evidence: ASR 431.9 ms and TTS first-byte 851.6 ms on the recorded comparison run.

Class mean/p95 cells above remain blank until the updated upstream direct-feed run is executed.

### CHALLENGES & REFERENCES

Pi audio: restore Fifine ALSA capture; run a human overlap/echo check.

Benchmark: run commit 57eaddb; attach stage marks and latency chart.

Telegram text is implemented; voice calls need Twilio or reviewed MTProto/TDLib.

Refs: MCP, Pipecat, Qwen vLLM, upstream fd-benchmark, ME3 repo/logs.